# Beyond Variance

## Asymmetric Labor Income Risk and Portfolio Entry

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Job Market Paper · October 2026

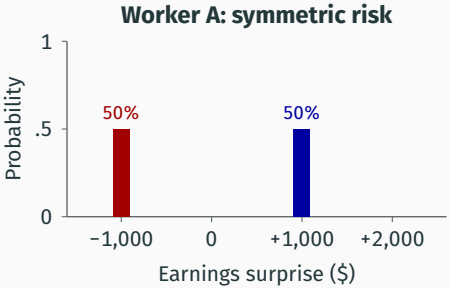
# Roadmap

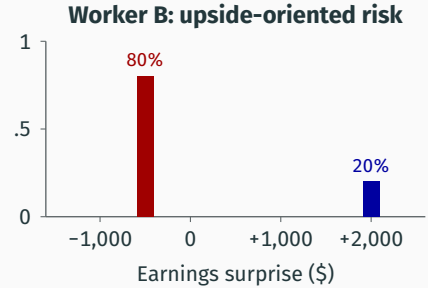
**1. Earnings risk**
Measurement and variation across workers

**2. Portfolio entry**
Earnings gains, losses and later ownership

**3. Life-cycle model**
Earnings paths and participation decisions

# Variance and earnings composition




Same mean surprise: \$0 Same variance: $\$1{,}000^2$

Stock market entry?

# Main findings

- **Annual entry: +1.60 pp** per source SD of composition
  Baseline entry rate: 8.82%

- **More upside predicts entry and later ownership**
  Entry followed by one or two later holdings

- **Model: opposing tail signs, smaller magnitudes**
  Earnings-risk coefficients excluded from estimation

# Contribution

**Empirical comparison**

- Historical CPS earnings distributions linked to later SIPP entry
- Earnings gains, losses and entry at given resources

**Quantitative interpretation**

- Entry estimates from a model fitted to financial outcomes
- Earnings paths and household decision rules

▶ Related evidence

# Income risk and stock market participation

**Earnings dynamics**

Guvenen, Ozkan & Song (2014)

Arellano, Blundell & Bonhomme (2017)

Guvenen et al. (2021)

**Income risk and portfolio choice**

Guiso, Jappelli & Terlizzese (1996)
Angerer & Lam (2009)
Fagereng, Guiso & Pistaferri (2018)
Catherine, Sodini & Zhang (2024)

**Participation and market access**

Fagereng, Gottlieb & Guiso (2017)
Briggs et al. (2021)
Gálvez (2023)
Galaasen & Raja (2024)
Bonaparte et al. (2025)

**This paper: earnings composition and entry**

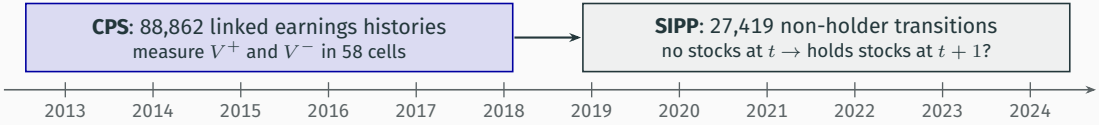
More upside predicts more entry.
More downside predicts less entry.

Entry and subsequent ownership at given mean growth, variance and observed resources.

**Consumption and saving**

Carroll (1997)
Gourinchas & Parker (2002)
Eberly (1994)

# Data and timing




- **CPS:** earnings distributions in 58 industry–education–age cells
- **SIPP:** historical risk assigned by baseline cell
- **Entry:** next-year stock/mutual-fund ownership among initial nonholders
  Outside retirement accounts

▶ Sample and units

# Earnings growth

**1. Annual nominal wage growth**

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}, \qquad y_{it} > 0, \quad y_{i,t+1} \geq 0.$$

$y_{it}$: annual wages of CPS worker $i$ in earnings year $t$. Zero destination wages: $g_{i,t+1} = -2$.

**2. Remove the industry–education age profile**

$$\epsilon_{i,t+1} = g_{i,t+1} - \left(\widehat{\theta}_{0G} + \widehat{\theta}_{1G} z_{it} + \widehat{\theta}_{2G} z_{it}^2 + \widehat{\theta}_{3G} z_{it}^3\right).$$

CPS-weighted cubic within industry–education group $G$.
$z_{it} = (a_{it} - 45)/20$; $a_{it}$ is age at the first interview.

**3. Center within the earnings cell**

$$\mu_c = \mathbb{E}_c[\epsilon], \qquad \text{within-cell deviation} = \epsilon_{i,t+1} - \mu_c.$$

$c$: industry–education–age-band cell.

# Upper- and lower-tail variation

Age-adjusted earnings growth, centered within each cell

$$V_c^- = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon < \mu_c\}], \qquad V_c^+ = \mathbb{E}_c[(\epsilon - \mu_c)^2 \mathbf{1}\{\epsilon > \mu_c\}].$$

**Total dispersion**

$$V_c = V_c^+ + V_c^-$$

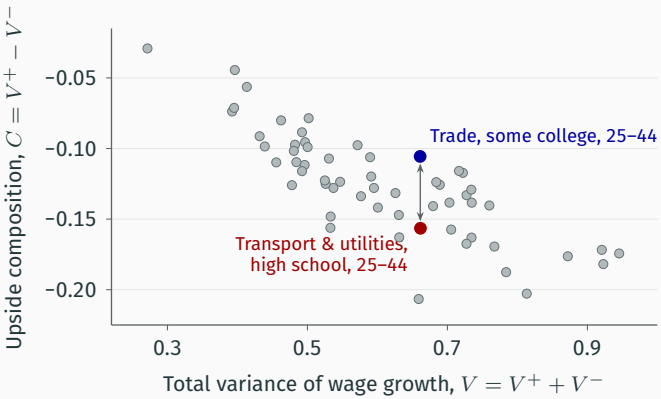
**Upside composition**

$$C_c = V_c^+ - V_c^-$$

Higher $C$ at fixed $V$: more upper-tail variation

▶ Growth construction ▶ Relation to skewness

# Variation across earnings cells




**58 earnings cells**

Highlighted pair:
Variance $\approx 0.66$
Composition gap $\approx 1.5$ source SD

All cells: $C < 0$
Relative upside measure

$\text{corr}(C, V) = -0.81.$

# Entry regression

$$H_{i,t+1} = \alpha + \beta_E \widetilde{C}^{\text{up}}_{c(i,t)} + \gamma_1 \widetilde{V}_{c(i,t)} + \gamma_2 \widetilde{\sqrt{V}}_{c(i,t)} + \gamma_3 \widetilde{\mu}_{c(i,t)} + \gamma_4 \big(\widetilde{\mu}_{c(i,t)}\big)^2$$
$$+ \delta_1 \log\big(\max\{A_{it}, 1\}\big) + \delta_2 \log\big(\max\{\bar{y}^{(1)}_{it} + 1, 1\}\big) + \delta_3 \mathbf{1}\big\{\min_{m \in \mathcal{M}_{it}} y^{(1)}_{itm} \leq 0\big\}$$
$$+ \delta_4 a_{it} + \delta_5 a^2_{it} + \delta_6 \text{Male}_{it} + \eta_{j(i,t)} + \kappa_{e(i,t)} + \lambda_{b(i,t)} + \tau_{t+1} + u_{i,t+1}, \quad H_{it} = 0.$$

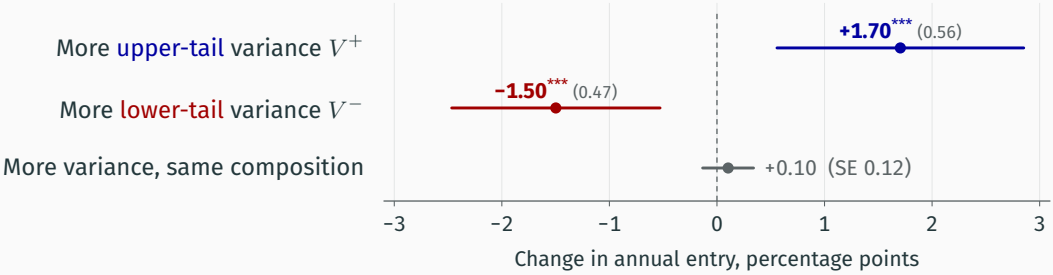
- **Outcome:** any-month stock/mutual-fund ownership next year
  Initial nonholders
- **Risk:** composition, dispersion and mean growth
  Fixed CPS source standardization
- **Controls:** assets, earnings, nonpositive earnings and demographics
  Industry, education, age-band and year fixed effects

▶ Inference ▶ Risk preferences

# Portfolio entry

| Growth measure | (1) Wages zeros included Arc | (2) Total earnings zeros included Arc | (3) Total earnings positive only Real log | (4) Total earnings positive only Arc |
|---|---|---|---|---|
| Upside composition | 1.601*** | 1.042** | 0.875*** | 0.695*** |
| | (0.506) | (0.405) | (0.279) | (0.248) |
| Conditional-SD magnitude (pp) | 0.590 | 0.506 | 0.573 | 0.534 |
| Composition reliability | 0.73 | 0.62 | 0.28 | 0.24 |
| Group-specific cubic age adjustment | Yes | Yes | Yes | Yes |
| Risk and baseline controls | Yes | Yes | Yes | Yes |
| Industry, education, age-band, year FE | Yes | Yes | Yes | Yes |
| Nonholder transitions | 27,419 | 27,419 | 27,419 | 27,419 |
| Mean annual entry (%) | 8.82 | 8.82 | 8.82 | 8.82 |
| Assignment cells / inference groups | 58 / 30 | 58 / 30 | 58 / 30 | 58 / 30 |

# Earnings gains, losses and entry




▶ Derivative algebra

# Alternative specifications

| Specification | Composition (pp) | SE |
|---|---|---|
| Prospective-sample baseline | 1.65** | 0.60 |
| Industry × year; education × year | 1.66** | 0.62 |
| Industry × education effects | 1.86** | 0.88 |
| Reported, self-reported, stable households | 2.56*** | 0.56 |
| Controls for realized earnings | 1.57** | 0.59 |

▶ Risk preferences and sorting

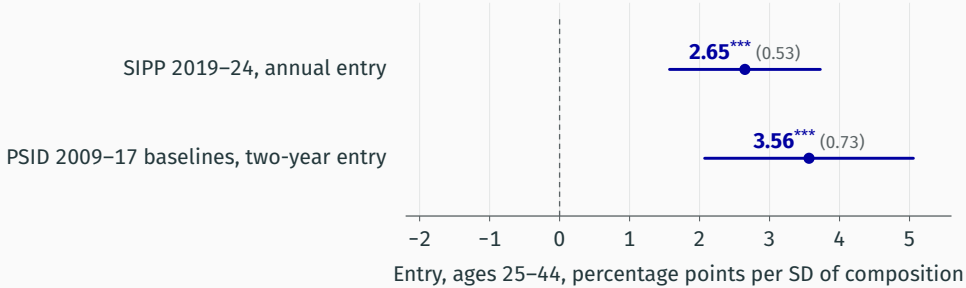
# Entry and subsequent ownership

| Joint outcome | Entry + one later holding | Entry + two later holdings |
|---|---|---|
| Composition estimate (pp) | 0.98* | 1.89** |
| Standard error | (0.50) | (0.79) |
| Initial nonholders | 9,236 | 3,254 |

**Outcome:** initial nonholder, then ownership at entry and later interviews

▶ Joint versus conditional survival

# Entry at ages 25–44




▶ Period gap and all-age results

# Retirement contributions

| Initial non-IRA stock ownership | Composition (pp) | SE |
|---|---|---|
| Nonholders | 0.22 | 0.33 |
| Holders | 6.36*** | 2.19 |
| Holder minus nonholder | 6.14** | 2.33 |

Outcome: any private IRA or annuity contribution

# Upside beliefs and risky reallocation

Survey of Consumer Expectations (SCE), retirement accounts

| Belief measure | Reallocation | Equity share |
|---|---|---|
| Upside-minus-downside tilt | $\mathbf{0.028}^{**}$ | $-0.576$ |
| | (0.013) | (0.664) |
| Sum of semideviations | $-0.017$ | $\mathbf{-3.805}^{***}$ |
| | (0.018) | (1.011) |
| Respondents | 2,136 | 3,332 |
| Controls | Expected earnings growth, age, education and income | |
| Fixed effects / SEs | State and module year / 50 state clusters | |

Reallocation score: $+1$ riskier, $-1$ safer, $0$ unchanged. Equity share: percentage points.

# Household expenditure

| Asinh dollar scale | Furnishings/ equipment $1,000 | Selected nondurables/ services $10,000 | Food $10,000 |
|---|---|---|---|
| *Wage composition, holding variance fixed* | | | |
| Initial nonholders | 0.007 | 0.001 | 0.002 |
| | (0.013) | (0.008) | (0.005) |
| Initial holders | 0.114*** | 0.012 | 0.009 |
| | (0.041) | (0.020) | (0.017) |
| Holder minus nonholder | 0.107** | 0.011 | 0.006 |
| | (0.046) | (0.024) | (0.019) |
| Baseline-weight-share average | 0.027** | 0.003 | 0.004 |
| | (0.012) | (0.006) | (0.004) |
| *Dispersion, holding composition fixed* | | | |
| Initial holders | 0.021*** | 0.003 | 0.001 |
| | (0.006) | (0.004) | (0.003) |
| 33,353 observations; 30 industry–education clusters | | | |

**Furnishings: stronger association among initial holders**

# Participation threshold

Log-utility example, earnings realized before the portfolio choice

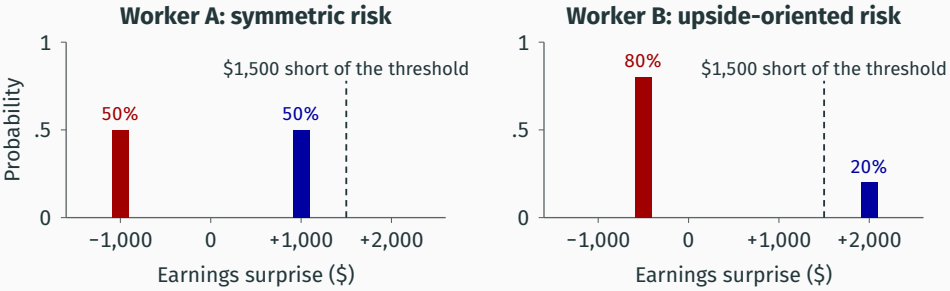
$$\text{Hold equity} \iff w + Y \geq w^*.$$

- Fixed cost relative to available resources
- Entry response concentrated near the threshold

Related margins: retirement contributions and durable purchases
Durable adjustment thresholds: Eberly (1994)

▶ Threshold formula

# Earnings gains near the threshold




Entry probability: **0%** for A **20%** for B

## Earnings risk and stock market participation

Saving, entry opportunities and holding costs

# Related models

| Published model | Model and finding | This paper adds |
|---|---|---|
| Briggs et al. (2021, JFE) | Life-cycle saving, entry and holding costs. Overpredicted entry after lottery windfalls. | Measured earnings environments. Intermittent entry opportunities and holding costs. |
| Catherine (2022, RFS) | Cyclical earnings skewness. Joint earnings and stock losses, low participation among the young. | Resource accumulation and entry. Independent stock and earnings shocks. |
| Gálvez & Paz-Pardo (2026, JFE) | Non-normal earnings shocks and nonlinear persistence in life-cycle portfolios. | Opposite entry patterns for upside and downside risk. Contributions of earnings paths and participation choices. |

▶ Related model mechanisms

# Participation decision

Observed before choice: resources, access and recurring cost

$$V_t(M, z, d, a, \xi) = \begin{cases} \mathcal{V}_t^0(M, z), & a = 0, \\ \max\{\mathcal{V}_t^0(M, z), \mathcal{V}_t^1(M - \kappa_R \xi, z)\}, & a = 1. \end{cases}$$

- Consumption and saving chosen in both branches.
- Equity share chosen when participating.
- Continuation values include future earnings, exit and re-entry.

▶ Recursive household problem

# Budget and portfolio constraints

$$c_t + A_t + D_t \kappa_R \xi_t = M_t,$$

$$M_{t+1} = A_t\{R_{f,t+1} + \alpha_t(R^s_{t+1} - R_{f,t+1})\} + Y_{t+1}.$$

- Participation: $D_t \in \{0, 1\}$, with $D_t \leq a_t$.
- $0 \leq \alpha_t \leq D_t$; no borrowing or short positions.
- Funded equity: $\alpha_t A_t \geq \$1 \times D_t$.

# Entry opportunities and holding costs

$$\Pr(a_t = 1 \mid D_{t-1}) = D_{t-1} + (1 - D_{t-1})q, \qquad q = 0.176.$$

$$K_t = \kappa_R \xi_t, \qquad \mathbb{E}[\xi_t] = 1, \qquad \kappa_R = \$524.$$

- Nonholders: a new opportunity draw each year.
- Holders: annual choice to continue or exit.
- Access permits entry; payment and investment remain choices.
- After exit: return to the nonholder opportunity process.

▶ Interpretation and selected payments ▶ Friction experiments ▶ Permanent alternative

# Earnings process

| Ingredient | Economic role | Empirical discipline |
|---|---|---|
| Persistent promotions and losses | Lasting earnings changes | CPS one-year and PSID multi-year moments |
| Transitory shocks and reporting error | Short-lived earnings variation | One-, two-, four- and six-year moments |
| Departures and returns to wage work | Employment and resource changes | Zero-wage spells |
| 58 cell adjustments | Growth, dispersion, asymmetry and departures | Industry–education–age environments |

Earnings laws are held fixed during financial estimation.

# Calibration and parameter estimates

## Calibration and data inputs

| Inputs held fixed | Value / source |
|---|---|
| Safe gross real return | 1.02 |
| Ordinary stock gross mean / SD | 1.06 / 0.18 |
| Annual crash probability | 1% |
| Crash net real return | $-48.5\%$ |
| EIS $\psi$ | 0.5 |
| Earnings laws / resources | CPS / PSID |
| Observed initial states | SIPP |

## Estimated financial parameters

| Parameter | Estimate |
|---|---|
| Risk aversion $\gamma$ | 8.801 |
| Discount factor $\beta$ | 0.811 |
| Recurring mean $\kappa_R$ | \$524 |
| Log-cost dispersion $\sigma_\kappa$ | 0.944 |
| Opportunity probability $q$ | 0.176 |

Joint simulated method of moments (SMM): five common parameters, eight financial targets.

**Calibrated:** returns and EIS. **Estimated separately:** earnings laws.
**From data:** resource profiles and initial household states.

▶ SMM targets and weighting ▶ Financial fit ▶ Criterion and numerical checks

# Untargeted entry regressions

17,606 observed initial nonholders; financial parameters held fixed

1. **Initial states:** each worker's observed resources and inferred earnings state.
2. **Entry probability:** average over earnings, opportunities and cost draws.
3. **Regression:** replace observed entry (0 or 1) with its predicted probability; retain regressors and survey weights.

$$\widehat{b}^{L} = (X'WX)^{-1}X'W\widehat{p}^{L}, \qquad L \in \{\text{cell}, \text{pooled}\}.$$

Predicted entry; same earnings-risk measures, controls, fixed effects and survey weights.

# Cell and pooled earnings laws

**Pooled benchmark**

- Education-specific career dynamics retained.
- Cell adjustments to growth, dispersion, asymmetry and departures removed.
- Optimal choices recomputed.

**Hold fixed**

- Financial parameters and observed origins.
- Initial earnings posteriors and group wage levels.
- Returns, regressors and survey weights.

$$\Delta b_C = b_C^{\text{cell}} - b_C^{\text{pooled}}.$$

▶ Interpretation

# Entry estimates: data and model

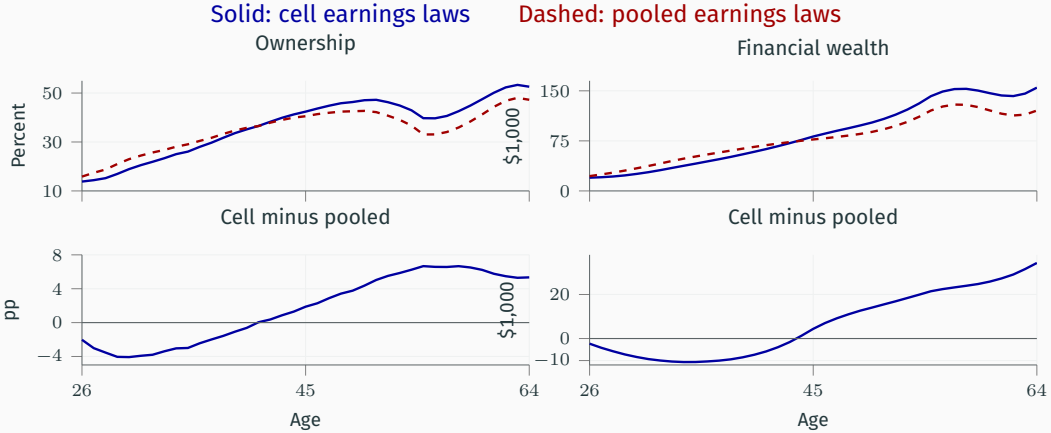
Opportunity model: untargeted projections on 17,606 matched origins

| Entry estimate (pp) | Data | SE | Cell | Pooled | Net |
|---|---|---|---|---|---|
| Composition | $1.816^{***}$ | (0.480) | 0.241 | -0.005 | 0.247 |
| Upper tail | $1.967^{***}$ | (0.513) | 0.115 | -0.116 | 0.230 |
| Lower tail | $-1.665^{***}$ | (0.470) | -0.368 | -0.105 | -0.263 |
| Dispersion | $0.151$ | (0.104) | -0.127 | -0.110 | -0.016 |

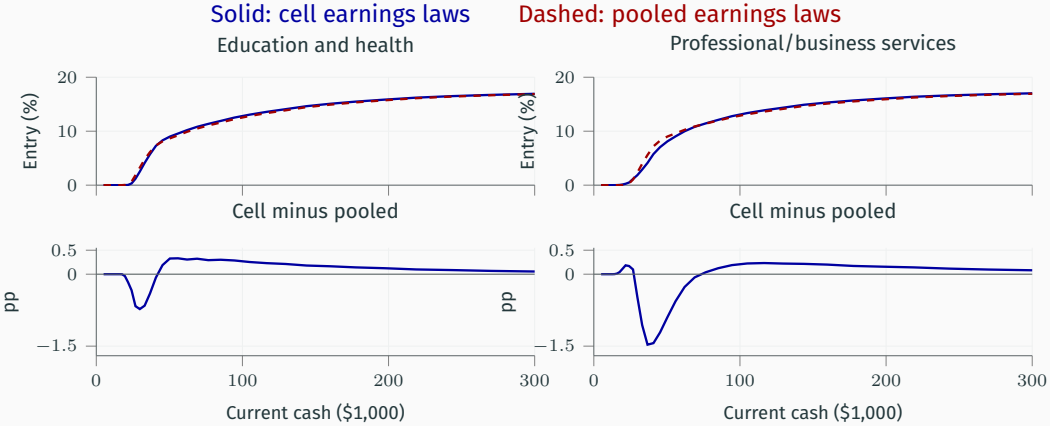
- Cell vs. pooled: upper-tail estimate turns positive, lower-tail estimate becomes more negative.
- Tail magnitudes well below the data.

▶ Law and policy contributions ▶ Coefficient figure ▶ Permanent alternative

# Ownership and wealth paths



# Entry probabilities by current cash



# Model results and limitations

- Cell laws reproduce opposing tail signs without fitting the coefficients.
- Earnings realizations affect entry through resources and saving.
- Recurring costs and repeated opportunities shape later ownership.
- Entry estimates too small, nonholder wealth growth too high.

▶ Windfall response ▶ Remaining discrepancies ▶ Permanent alternative

# Conclusion

- **Entry: +1.60 pp** per source SD of earnings composition
  More upside predicts more entry, more downside predicts less

- **Subsequent ownership:** also more likely with more upside

- **Life-cycle model:** matching signs, smaller magnitudes
  Most of the model association comes from earnings paths

▶ Discussion appendix

# Questions and discussion

Tai Lo Yeung

▶ Discussion appendix

# Discussion appendix

**Evidence and measurement**

- ▶ Sample
- ▶ Earnings construction
- ▶ Skewness
- ▶ Composition example
- ▶ Four constructions
- ▶ Entry results
- ▶ Inference
- ▶ Tail algebra
- ▶ Risk preferences
- ▶ Sustained entry
- ▶ External evidence
- ▶ Holder outcomes
- ▶ Entry and retention
- ▶ Consumption mechanism

**Model and financial fit**

- ▶ Financial fit
- ▶ Risk coefficient figure
- ▶ Model ingredients
- ▶ Observed initial states
- ▶ Life-cycle profiles
- ▶ Sustained model ownership
- ▶ Parameters
- ▶ Targets and weighting
- ▶ Costs and opportunities
- ▶ Financial wealth
- ▶ Pooled laws
- ▶ Law and policy contributions
- ▶ Later ownership estimates
- ▶ Recursive problem

**Comparisons and experiments**

- ▶ Portfolio-model literature
- ▶ Analytical identities
- ▶ Related empirical evidence
- ▶ Selected references
- ▶ Permanent eligibility
- ▶ Alternative financial fit
- ▶ Alternative entry estimates
- ▶ Alternative dynamics
- ▶ Participation frictions
- ▶ Earnings news
- ▶ Windfall response
- ▶ Model mechanisms
- ▶ Model assessment
- ▶ Numerical checks
- ▶ Experiment coverage

◀ Back

# Targeted financial moments

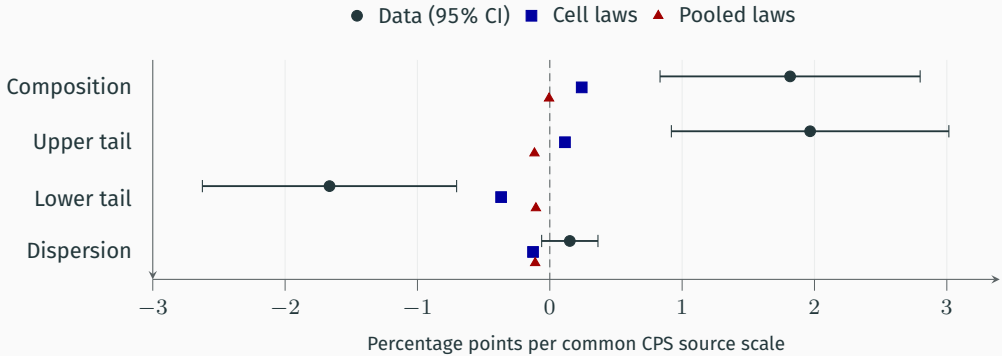
| Moment | Unit | Data | SE | Model |
|---|---|---|---|---|
| Entry: low initial wealth | % | 2.93 | (0.26) | 3.18 |
| Entry: middle initial wealth | % | 5.84 | (0.36) | 5.51 |
| Entry: high initial wealth | % | 10.36 | (0.48) | 10.44 |
| Retention: low initial equity share | % | 66.19 | (1.43) | 67.75 |
| Retention: high initial equity share | % | 76.31 | (1.23) | 74.34 |
| Next equity share: all initial holders | % | 33.38 | (0.84) | 34.40 |
| Wealth change: initial nonholders | $ | 3,939 | (778) | 9,941 |
| Wealth change: initial holders | $ | 1,946 | (5,177) | 511 |

Main discrepancy: excess nonholder wealth growth

◀ Back ▶ Additional diagnostics ▶ Alternative fit

# Entry estimates from the model

17,606 matched initial nonholders, untargeted coefficients




◀ Back ▶ Permanent alternative

# Earnings risk and portfolio models

**Non-Gaussian earnings risk**

Barndorff-Nielsen, Kinnebrock & Shephard (2010)
Segal, Shaliastovich & Yaron (2015)
Ghysels, Plazzi & Valkanov (2016)
Arellano, Blundell & Bonhomme (2017)
Guvenen et al. (2021)

**Life-cycle portfolio models**

Cocco, Gomes & Maenhout (2005)
Gomes & Michaelides (2005)
Catherine (2022)
Gálvez & Paz-Pardo (2026)

**This paper**

**Measurement.** Upper and lower partial variances from zero-inclusive earnings histories, measured independently of portfolio choices.

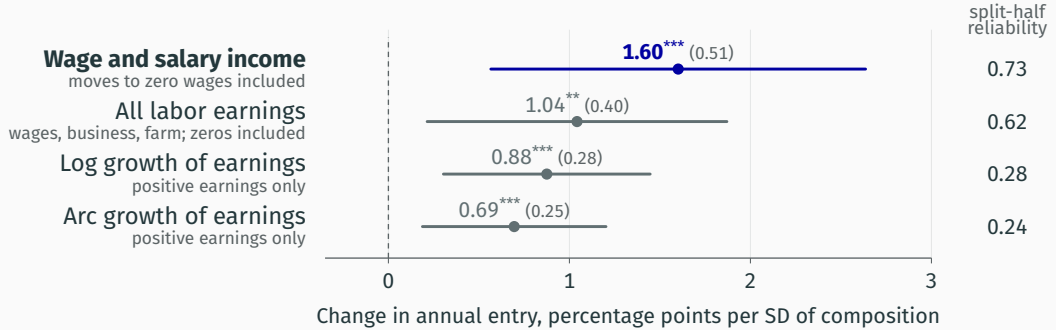
**Untargeted predictions.** Cell earnings laws predict more entry with upside risk and less with downside risk. Pooled laws miss this pattern in the main model.

# Composition at fixed variance

| | Symmetric risk | Upside-oriented risk |
|---|---|---|
| Lower outcome | $-1{,}000$ (50%) | $-500$ (80%) |
| Upper outcome | $+1{,}000$ (50%) | $+2{,}000$ (20%) |
| Lower partial variance | 500,000 | 200,000 |
| Upper partial variance | 500,000 | 800,000 |
| Total variance | 1,000,000 | 1,000,000 |
| **Composition** | **0** | **600,000** |

**Composition: upper minus lower partial variance.**

# Entry estimates across earnings measures




**Positive entry estimates across all four measures. Zero-inclusive wages are the most reliable.**

▶ Zeros and growth ▶ Skewness

# Entry and retention

**Established owners**

Resources well above the threshold

Participation under either earnings law

**Marginal entrants**

Entry after a favorable earnings realization

Later ownership depends on subsequent resources

**Lower average retention need not imply lower retention for every incumbent.**

▶ Holder outcomes ▶ Probability identities

# Model ingredients

| Model ingredient | Economic role |
|---|---|
| Career earnings risk | Realized income and future resources |
| Optimal saving | Resources carried forward |
| Entry opportunity | Chance to consider a funded equity position |
| Recurring cost | Ownership when benefits exceed the holding cost |

# Observed initial states

21,188 SIPP worker-year pairs, ages 25–63 at origin

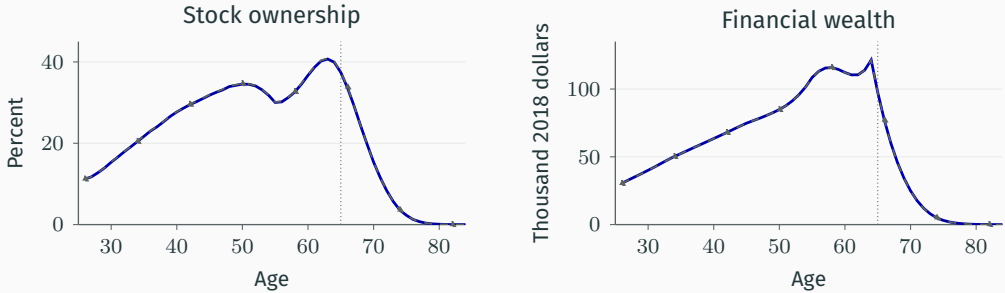
| Input | Use in the model |
|---|---|
| Wages, wealth, ownership and shares | Latent earnings from wages; observed financial positions |
| Taxes and own pension contributions | Wages net of taxes and contributions |
| Own retirement income | PSID Social Security and pensions by education and age |
| Calendar returns | Returns on inherited portfolios in observed transitions |

Future return expectations: stationary.

# Conditional life-cycle profiles

Conditional projection from 452 observed age-25 origins in 29 cells

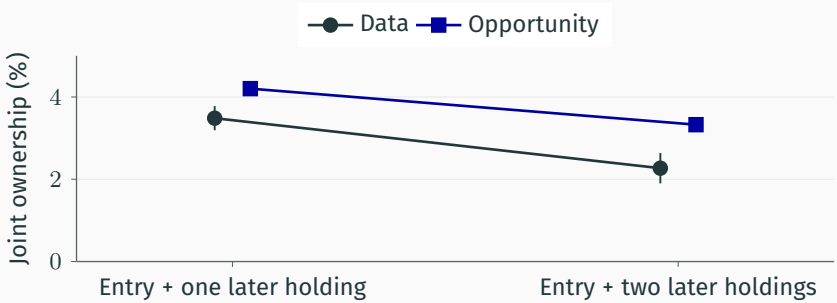
Solid: 141-node grid  Dashed triangles: 201-node grid




**Numerical accuracy:** almost identical paths on finer cash and return grids.

# Entry and subsequent ownership

Untargeted predictions on common data/model followup cohorts




Sustained ownership levels remain overpredicted.

▶ Common-cohort estimates

# Samples

| Sample | Observations |
|---|---|
| CPS linked earnings histories | 88,862 |
| CPS histories with zero destination wages | 5,490 |
| SIPP full entry sample | 27,419 |
| SIPP prospective allocation sample | 31,569 |
| Model financial-fitting worker-year pairs | 21,188 |
| Matched model/data entry regression | 17,606 |

Worker transitions, household assets. Mixed-asset mutual funds included. Retirement-account allocation unobserved.

# Arc wage growth

$$g_{i,t+1} = \frac{2(y_{i,t+1} - y_{it})}{y_{i,t+1} + y_{it}}.$$

- Positive origin wages, zero destinations included: $g = -2$
- Weighted cubic age adjustment by industry–education group
- Residuals centered within industry–education–age cells
- Zero wages: nonemployment or a different income source, not necessarily dismissal

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# Composition and skewness

| Zero-inclusive wage measure | Composition | Skewness |
|---|---|---|
| Split-half reliability | 0.73 | 0.39 |

- Partial correlation given $V, \sqrt{V}$: 0.91
- Composition: difference between upper and lower partial variances
- Substantial overlap with conventional skewness

# Alternative earnings measures

| Construction | Composition (pp) | SE |
|---|---|---|
| Wages, zeros included | 1.601*** | 0.506 |
| Total labor earnings, zeros included | 1.042** | 0.405 |
| Real log growth, positive support | 0.875*** | 0.279 |
| Nominal arc growth, positive support | 0.695*** | 0.248 |

Controls: $V, \sqrt{V}, \mu, \mu^2$, household characteristics
Industry, education, age-band and year fixed effects

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# Statistical inference

- **Coefficient tests:** 30 industry–education clusters, CR1/$t_{29}$
- **Wild-cluster bootstrap:** $p = 0.039$
- **Source uncertainty:** re-estimated CPS moments in separate checks
- Restricted samples: actual cluster count. Unadjusted coefficient stars.

# Tail derivatives

Risk component in unstandardized units

$$f(V^+, V^-) = b_C(V^+ - V^-) + b_V(V^+ + V^-) + b_S\sqrt{V^+ + V^-}.$$

$$\frac{\partial f}{\partial V^+} = b_C + b_V + \frac{b_S}{2\sqrt{V}}, \qquad \frac{\partial f}{\partial V^-} = -b_C + b_V + \frac{b_S}{2\sqrt{V}}.$$

- Half-difference: composition coefficient
- Half-sum: local dispersion derivative
- Common reference volatility and CPS source scaling

# Risk preferences and sorting

- Risk preferences affect participation thresholds and earnings responses
- Composition slopes may differ across preference groups
- Reported risk appetite: noisy sorting diagnostic

Sorting remains a possible source of the association.

# Entry and subsequent ownership

| Composition estimate (pp) | One followup | Two followups |
|---|---|---|
| Any entry | 2.09*** | 3.31** |
| Entry and later holdings | 0.98* | 1.89** |
| Entry with an interruption | 1.11** | 1.42 |
| Initial nonholders | 9,236 | 3,254 |

Joint outcomes partition entry within each cohort.
Imprecise estimates of retention among entrants. Followup selection and weaker three-date reporting screen.

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# Period differences and age groups

| Young-worker SIPP entry gap (pp) | Estimate |
|---|---|
| Original period gap | 2.885 |
| After symmetric overlap weighting | 2.623 |
| Change in gap | $-0.262$ |
| SE of change | 0.294 |
| $p$-value of change | 0.381 |

PSID composition: all ages $-0.18$ (SE 0.39)
Ages 45–65: 0.44 (SE 1.30). All-age replication inconclusive.

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# Initial-holder outcomes

| Initial-holder outcome | Composition (pp) | SE |
|---|---|---|
| Next-period ownership / retention | $-3.72^{**}$ | 1.71 |
| Next risky share, controlling initial share | $-0.51$ | 0.94 |

- Retention wild-cluster bootstrap $p = 0.12$.
- All initial holders, including subsequent exits
- Initial-holder retention differs from survival among new entrants

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# Participation and consumption

Auxiliary two-date log-utility example with endogenous saving

$$c^*(m, d) = \frac{m - \kappa d}{1 + \beta}.$$

$$\Delta \mathbb{E}[c] = -\frac{\kappa}{1 + \beta} \Delta \Pr(d = 1) \qquad \text{at fixed mean resources.}$$

- Small consumption response possible, exact invariance unnecessary
- IRA/annuity outcome: contribution participation only
- Furnishings: positive primary asinh estimate among holders. Imprecise dollar and original-construction estimates.

# Threshold and selection formulas

$$w^* = \frac{\kappa}{1 - e^{-b}} - h, \qquad \pi(w, F) = \Pr(Y \geq w^* - w).$$

$$\text{Entry} = \frac{\mathbb{E}[\pi(1-\pi)]}{\mathbb{E}[1-\pi]}, \qquad \text{Retention} = \frac{\mathbb{E}[\pi^2]}{\mathbb{E}[\pi]},$$

$$\text{Entry + } k \text{ later holdings} = \frac{\mathbb{E}[(1-\pi)\pi^{k+1}]}{\mathbb{E}[1-\pi]}.$$

$b > 0$: optimized log-return gain. $h$: safe nontradable terminal income.
Fixed wealth, independent draws across dates. Signs depend on resources and earnings laws.

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# Selected references


- Catherine, S. (2022). Countercyclical Labor Income Risk and Portfolio Choices over the Life Cycle. *Review of Financial Studies*.
- Catherine, S., P. Sodini and Y. Zhang (2024). Countercyclical Income Risk and Portfolio Choices: Evidence from Sweden. *Journal of Finance*.
- Briggs, J., D. Cesarini, E. Lindqvist and R. Östling (2021). Windfall Gains and Stock Market Participation. *Journal of Financial Economics*.

# Related empirical evidence

| Study | Empirical comparison | This paper's contribution |
|---|---|---|
| Gálvez (2023) | Participation responses to persistent earnings shocks | Historical earnings environments before subsequent entry |
| Catherine, Sodini and Zhang (2024) | Cyclical tail exposure and stockholding | Opposite tail estimates. Composition at fixed variance. |
| Van den Akker et al. (2026)* | Subjective income expectations linked to administrative portfolios | Independently measured environments, entry and sustained ownership |

Complementary subjective-expectations evidence

# Permanent participation eligibility

$$T_i \in \{0, 1\}, \qquad T_i \text{ remains fixed throughout the path.}$$

- Current holders and reported former holders: $T_i = 1$.
- Among other initial nonholders, $\Pr(T_i = 1) = 0.119$.
- Eligible: annual choice to pay and hold equity. Ineligible: safe asset only.
- Five parameters fitted separately to the same eight financial targets.

Permanent exclusion applies to a fixed share of initially unknown nonholders.

# Financial fit under both access models

| Moment | Data | Opportunity | Permanent |
|---|---|---|---|
| Entry: low initial wealth | 2.93 | 3.18 | 3.21 |
| Entry: middle initial wealth | 5.84 | 5.51 | 5.49 |
| Entry: high initial wealth | 10.36 | 10.44 | 10.39 |
| Retention: low initial equity share | 66.19 | 67.75 | 68.01 |
| Retention: high initial equity share | 76.31 | 74.34 | 73.83 |
| Next equity share: all initial holders | 33.38 | 34.40 | 33.81 |
| Wealth change: initial nonholders | 3,939 | 9,941 | 10,005 |
| Wealth change: initial holders | 1,946 | 511 | 501 |

Both models: positive wealth growth, excessive nonholder accumulation.

# Entry estimates under permanent eligibility

| Entry estimate (pp) | Data | Opportunity net | Permanent net |
|---|---|---|---|
| Composition | 1.816 | 0.247 | 0.171 |
| Upper tail | 1.967 | 0.230 | 0.155 |
| Lower tail | -1.665 | -0.263 | -0.188 |
| Dispersion | 0.151 | -0.016 | -0.016 |

- Pooled laws already give opposite signs. Both estimates move further from zero with cell laws.
- Net magnitudes remain well below the data in both models.

# Participation dynamics under both access models

| Untargeted outcome | Data | Opportunity | Permanent |
|---|---|---|---|
| Re-entry: reported former holders (%) | 27.16 | 9.83 | 57.68 |
| Entry + one later holding (%) | 3.48 | 4.20 | 3.82 |
| Entry + two later holdings (%) | 2.27 | 3.33 | 2.24 |
| Projected age-45 ownership (%) | — | 31.83 | 16.31 |
| Windfall response after two years (pp) | — | 15.25 | 3.44 |

Permanent eligibility: closer fit to longer sustained ownership; excessive re-entry among reported former holders.

# Parameter estimates by access model

| | Opportunity | Permanent |
|---|---|---|
| Risk aversion $\gamma$ | 8.801 | 8.917 |
| Discount factor $\beta$ | 0.811 | 0.810 |
| Recurring mean $\kappa_R$ | \$524 | \$413 |
| Log-cost dispersion $\sigma_\kappa$ | 0.944 | 1.013 |
| Opportunity $q$ / active share $p$ | 0.176 | 0.119 |
| EIS $\psi$, fixed | 0.5 | 0.5 |
| Additional entry fee | 0 | 0 |

Parameters fixed before earnings-risk and dynamic comparisons. No binding bounds.

# Estimation criterion and grid refinement

$$Q = Q_E + 0.1Q_H + Q_W = 9.77.$$

- Marginal covariance blocks: entry, holder outcomes and wealth changes.
- Wealth covariance: \$2,000 regularization. Primary-grid growth buffer: \$500 for both groups.
- Native convergence. At 201/7: wealth growth of \$9,937 for nonholders and \$498 for holders.
- Excluded targets: earnings risk, duration, windfalls and life-cycle profiles.

◀ Back ▶ Other numerical diagnostics ▶ Index

# Opportunity friction and recurring payments

- Opportunity: a nonholder can reconsider participation.
- Advice, attention and search are possible interpretations, not separately identified.
- Recurring cost paid only when holding equity.
- Offered cost: mean $524, median $335. Participants select lower draws.

| Selected recurring payment | Real 2018 dollars |
|---|---|
| Entrants, supported origins | 235 |
| Retained holders | 359 |

# Wealth growth and saving net of returns

$$\Delta A_t = \text{gains on inherited assets} + Y_t - c_t - D_t \kappa_R \xi_t.$$

| Nonholder wage tercile | Data growth ($) | Model growth ($) |
|---|---|---|
| Low | 2,388 | 3,588 |
| Middle | 236 | 8,705 |
| High | 9,195 | 17,535 |

Holders have positive wealth growth but negative aggregate saving net of returns.

# Pooled laws and initial-state sorting

- Fixed: origins, group wage levels, earnings posteriors and calendar returns.
- Remove cell adjustments; reoptimize at common financial parameters.
- Initial resources may remain correlated with earnings cells.
- Pooled holder wealth growth: about $483 below zero. Positive growth is imposed in the fitted cell model.

# Four-corner accounting by risk measure

$$\Delta_L = \tfrac{1}{2}\{B(c,c) - B(p,c) + B(c,p) - B(p,p)\},$$
$$\Delta_P = \tfrac{1}{2}\{B(c,c) - B(c,p) + B(p,c) - B(p,p)\}.$$

| Entry estimate (pp) | Laws | Decisions | Total |
|---|---|---|---|
| Composition | +0.274 | -0.027 | +0.247 |
| Upper tail | +0.302 | -0.072 | +0.230 |
| Lower tail | -0.245 | -0.018 | -0.263 |
| Dispersion | +0.029 | -0.045 | -0.016 |

Decision rules dampen the upper-tail contribution and reinforce the negative lower-tail contribution.

# Entry and later ownership: estimates

| | One later holding | Two later holdings |
|---|---|---|
| Origins | 5,459 | 1,801 |
| Data composition (pp) | 0.11 | 0.77 |
| Data SE | (0.66) | (0.77) |
| Opportunity net, two seeds | 0.19–0.20 | 0.18–0.22 |
| Permanent net, two seeds | 0.14–0.19 | 0.08–0.13 |

Smaller common cohorts: imprecise empirical composition estimates. Simulation-seed ranges are not confidence intervals.

# Model fit and remaining discrepancies

| Comparison | Model result | Discrepancy |
|---|---|---|
| Entry by wealth | Increasing across wealth terciles | Imperfect financial fit |
| Entry and earnings risk | Opposing net tail signs | Magnitudes well below data |
| Participation dynamics | Entry rises after gains and windfalls | Sustained levels too high; re-entry too low |
| Saving and positions | Positive wealth changes for both groups | Excess nonholder accumulation; too few small accounts |

Permanent alternative: excessive re-entry; closer sustained-ownership fit.

# Earnings-risk mechanisms in related models

| | Earnings mechanism | Portfolio implication |
|---|---|---|
| Catherine (2022) | Downside earnings risk when stock returns are poor | Joint downside exposure discourages equity |
| Gálvez–Paz-Pardo (2026) | Non-normal shocks and nonlinear persistence by age and earnings position | Lifetime earnings risk limits portfolio risk taking |
| This paper | Earnings environments and resources near participation margins | Opposite entry patterns for upside and downside risk, without fitting them |

Common framework: life-cycle saving, portfolio choice and participation frictions.

# Recursive household problem

$\rho = 1 - 1/\psi$, $\omega = 1 - \gamma$; $x$: resources after any recurring payment

$$V_{jt}(z,x) = \max_{(A,\alpha)\in\mathcal{A}_j(x)} \left[(x-A)^\rho + \beta s_t \left\{\mathbb{E}\left[\bar{V}_{t+1}(z', j, AR(\alpha) + Y')^\omega \mid z\right]\right\}^{\rho/\omega}\right]^{1/\rho}.$$

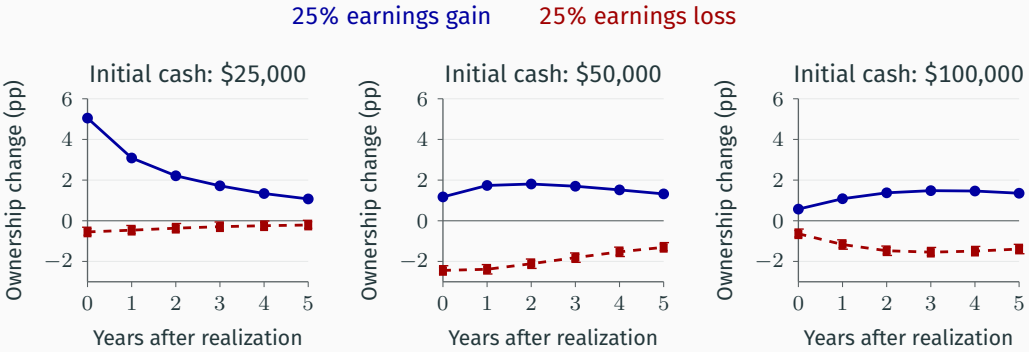
$$W_t(z,m,\xi) = \max\{V_{0t}(z,m), V_{1t}(z, m - \kappa_R\xi)\}.$$

$$\bar{V}_t(z,d,m)^\omega = (1-q_d)V_{0t}(z,m)^\omega + q_d\mathbb{E}_\xi[W_t(z,m,\xi)^\omega], \quad q_d = d + (1-d)q.$$

$\mathcal{A}_j$ imposes nonnegative assets and funded equity if $j = 1$. At age 84, $\bar{V}_{84}(z,d,m) = m$; no bequest motive.

# Response to realized earnings shocks

Age 45: a one-year shock equal to 25% of a fixed reference gross wage




▶ Earnings news at fixed current resources ◀ Back

# Realized earnings and earnings news

| Experiment | What changes | Interpretation |
|---|---|---|
| Realized transitory shock | Current cash; fixed future laws | Gains can finance entry |
| Upper-tail news | Future distribution; fixed current resources | Participation response can be small |
| Composition news | Future distribution; fixed current resources | Response varies with cash and horizon |

Composition-news responses can reverse sign at low cash. More upside need not raise entry.

# Numerical precision

- Financial fit, matched entry projections and four-corner accounting: declared finer-grid checks passed.
- Life-cycle grids: 141/5 and 201/7. Some local policies, cash responses and consumption remain grid-sensitive.
- Realized-earnings events: separate 301-node refinement.
- Simulation dispersion and grid differences measure numerical, not sampling, uncertainty.

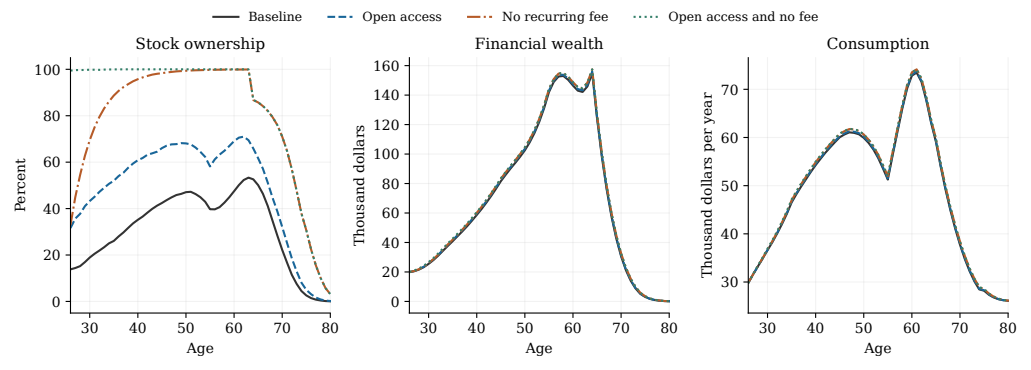
# Additional model experiments

| Family | Exercises in both specifications |
|---|---|
| Saving and portfolios | Age profiles, consumption, funded shares, entry and exit |
| Access frictions | Open access, zero and altered recurring costs |
| Earnings and retirement | Transitory and persistent risk, income insurance, pensions |
| Preferences and returns | Risk aversion, patience, EIS, expected returns and volatility |
| Household responses | Realized earnings, windfalls, fixed-resource news, value of future income |

39 scenarios per model, plus conditional policy and valuation diagnostics. Alternative and sensitivity results in the online appendix.

# Participation-friction counterfactuals

Remove each friction; reoptimize household choices




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# SMM targets and weighting

$$\theta = (\gamma, \beta, \kappa_R, \sigma_\kappa, q), \qquad \widehat{\theta} = \arg\min_{\theta} \{Q_E(\theta) + 0.1 Q_H(\theta) + Q_W(\theta)\}.$$

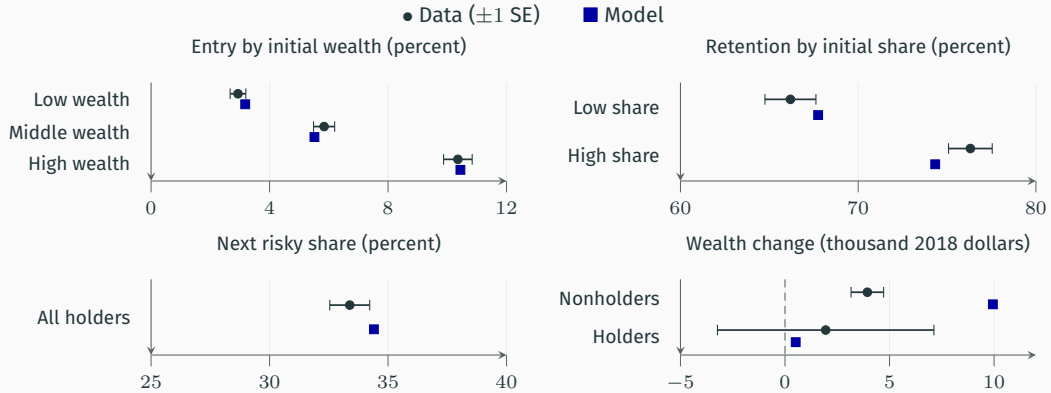
| Block | Targets | Economic discipline |
|---|---|---|
| Entry (3), $Q_E$ | Three rates by initial wealth | Opportunities and affordability |
| Holders (3), $Q_H$ | Two retention rates and one next equity share | Continued participation and allocation |
| Wealth (2), $Q_W$ | Two changes by initial ownership | Saving and resource use |

**Weight 0.1:** entry and wealth growth receive priority over retention and share fit.

◀ Back ▶ Financial fit ▶ All eight targets ▶ Exact criterion

# Financial targets: data and model

Eight financial moments; five estimated parameters




Main discrepancy: excessive wealth growth among initial nonholders.

◀ Back ▶ Full target table ▶ Index

# Earnings-law and policy contributions

**1. Earnings laws:** change future earnings and resources; retain the same choices at a given household state.

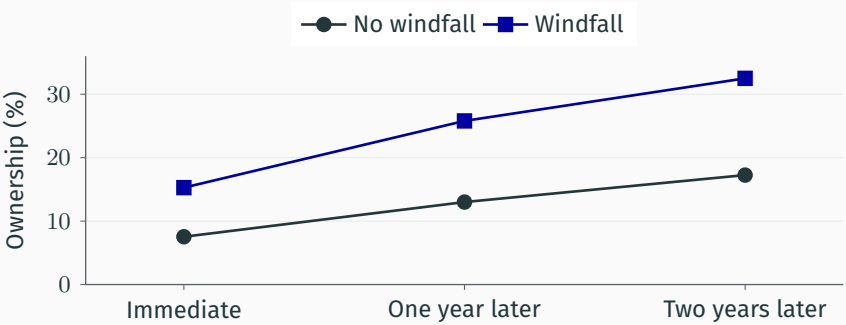
**2. Decision rules:** reoptimize saving, participation and portfolio shares under the new laws.

| Contribution to the composition estimate | pp per source SD |
|---|---|
| Earnings laws and resulting resources | $+0.274$ |
| Reoptimized decision rules | $-0.027$ |
| Total: cell minus pooled | $+0.247$ |

◀ Back ▶ Tail accounting and identities ▶ Index

# Windfall response

Untargeted experiment: $150,000 to 423 age-44 initial nonholders




**Model:** $+7.7$ pp immediately; $+15.2$ pp after two years.
**Swedish lottery evidence:** about $+12$ pp in the winning year (Briggs et al., 2021).

◀ Back ▶ Realized earnings gains ▶ Index